Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/23/2024
PI Name:	Alwood, Joshua Ph.D.		
Project Title:	VNSCOR: Oxidative Stress and the Neuroconsequences of Spaceflight Environment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-1490
Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	03/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	14	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2026 per L. Juliette/HFBP (Ed., 10/27/25). NOTE: End date changed to 9/30/2027 per L. Juliette/HFBP (Ed., 2/21/25).		
Key Personnel Changes/Previous PI:	PI change to Joshua Alwood, as April Ronca retired from NASA. NOTE: For prior reporting, see "VNSCOR: Oxidative Stress and the Neuroconsequences of Spaceflight Environment" under Dr. April Ronca (Ed., 11/24/24).		
COI Name (Institution):	Tahimic, Candice Ph.D. (University of North Florida) Paul, Amber Ph.D. (Embry-Riddle Aeronautical University) Mhatre, Siddhita Ph.D. (NASA Ames Research Center) Iyer, Janani Ph.D. (NASA Ames Research Center) Puukila, Stephanie Ph.D. (Blue Marble Space Institute of Science)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The proposed project will test the hypothesis that Ionizing Radiation (IR), microgravity, and social isolation combine synergistically to trigger an oxidative stress response that alters immune homeostasis, brain structure and function, and neurobehavioral and cognitive performance. Specific Aims for this project are: (1) Determine dose-response curves for acute 'Five-Ion GCR (galactic cosmic ray) Simulation' exposure for immune, brain, and performance responses in crew age-matched adult male and female mice; (2) Determine effects of acute 'Five-Ion GCR Simulation' exposure singly and in combination with simulated microgravity and social isolation, on immune, brain, and performance responses in crew age-matched male and female mice mimicking deep space missions; and (3) Determine efficacy of the dietary antioxidant, Nicotinamide Mononucleotide (NMN), a key intermediate in nicotinamide adenine dinucleotide (NAD+) biosynthesis. The project relies on established and highly translatable ground-based mouse models and assays with IR exposures to be performed at the NASA Space Radiation Laboratory (NSRL). The experimental approach will provide definitive data on the timing and mechanisms involved in the oxidative stress response, immune, and brain changes, and ensuing functional (behavioral/cognitive) impairments expected during human transit to Mars. This project will identify potential immune biomarkers for, and mechanisms underlying, structural and functional changes in the immune and nervous systems leading to behavioral/cognitive performance deficits, and its potential application to develop effective countermeasures to mitigate negative health effects of long duration space habitation. This proposal addresses NASA's efforts to rapidly advance the characterization of risks and identifying appropriate countermeasures in anticipation of future deep space missions. Ensuring crew health and performance during extended transits necessitates that sensorimotor and cognitive abilities remain strong to avoid potentially catastrophic health and safety outcomes. Further, despite historically low numbers of females astronauts, recent NASA Astronaut Corps class selections, comprised of 50% and 40% women as compared to men, signal the need to understand how sex differences affect physiological adaptation and health in the space environment. This integrative project, developed by a cross-disciplinary team highly experienced in spaceflight and radiation research, utilizes established space biosciences research protocols and variables, and time-honored, as well as modern, research methodologies. We will address major risks and associated gaps: (1) Risk of Acute (In-flight) and Late Central Nervous System Effects from Radiation (CNS), (2) Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks (SM), and Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders that have been combined into the NASA CNS, BMed, and SM (CBS) Integrated Research Plan. Here we have adhered to primary goals set forth in Human Exploration Research Opportunities (HERO) Appendix A to determine how key features of the deep space environment may interact to increase risk to a crew by negatively impacting health and performance, and we have identifed and developed strategies to characterize and mitigate the potential risks via countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Extensive progress was made during this reporting period on the first two project aims, with two publications and multiple abstracts focused on acute Galactic Cosmic Ray simulation (GCRsim) dose-response and combinatorial effects of spaceflight stressors (isolation, simulated weightlessness, and simulated galactic cosmic rays using GCRsim) on crew-aged mice. Data were collected on neural, behavioral, and immune outcomes at varying timepoints in relation to spaceflight stressor exposure. The Virtual NASA Specialized Center of Research (VNSCOR). The VNSCOR Team met extensively to discuss touchpoints across the project and hone our experimental designs. The VNSCOR includes two species, mouse and rat, with mature male and female subjects approximating crew ages, and six standardization protocols across three laboratories. The timeline for the combined space stress model remains similar across groups, with all animals experiencing each stressor for identical lengths of time and at the same timepoints – this will allow a standard model at the NASA Space Radiation Laboratory (NSRL) to be developed for future studies. According to this timeline, all Principal Investigator (PI) teams will perform hindlimb unloading as the weightlessness model, and irradiate all animals with the 5-ion GCRsim at both a low (15 cGy) and high dose (50 cGy) that most closely mimics the mission travel time to Mars. A slightly lower protein diet, containing no phytoestrogens, will be used to closely mimic the diet of astronauts and limit the effects of circulating estrogen. Additionally, efforts are being made to create similar animal care and handling conditions. To maintain all original grant aims, each group will return to their respective institutions and perform a minimum of two identical behavioral assays to measure changes in sensorimotor (adhesive removal) and memory (novel object recognition) functions. Using three Luminex-based arrays, each group will evaluate peripheral blood markers to elucidate potential predictive biomarkers of behavioral performance. These data, together, will be combined into a predictive statistical model to identify potential biomarkers that may predict behavior decrements and neuroinflammation across three institutions and two species. Predictive modeling efforts are underway, integrating multiple measures derived from the current project with planned expansions across VNSCOR Teams.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Mhatre SD, Iyer JS, Tabares Ruiz S, Puukila S, Hayes J, Paul AM, Tahimic CGT, Shirazi-Fard Y, Sowa M, Alwood JS, Ronca AE. "Neurobiological outcomes of mice exposed to combined spaceflight stressors." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Ronca AE, Mhatre SD, Iyer JS, Puukila S, Tabares Ruiz S, Hayes J, Brekker A, Burke M, Semel M, Lowe M, Rubinstein L, Sowa M, Shirazi Y, Tahimic CGT, Alwood JS, Paul AM. "Plasma and brain cytokine responses to single and combinatorial spaceflight stressors in male and gemale mice." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024

Abstracts for Journals and Proceedings	Paul AM, Cebula A, Burke M, Larsen E, Juarez K, Mitchell C, Juran CM, Tahimic CGT, Tabares Ruiz S, Iyer J, Puukila S, Mhatre SD, Alwood JS, Lowe M, Rubinstein L, Shirazi-Fard Y, Sowa MB, Ronca AE. "Immune mobilization circadian patterns in the spaceflight environment." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
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Articles in Peer-reviewed Journals	Brekker MA, Semel M, Korostenskij I, Krikourian O, Kelly KMW, Santos J, Lowe M, Mhatre S, Iyer J, Paul AM, Puukila S, Alwood JS, Shirazi-Fard Y, Rubinstein L, Goukassian D, Ronca AE, Tahimic CGT. "Simulated space radiation exposure leads to long-term changes in molecular signatures of the heart." Life Sci Space Res. 2026 May 2. Online ahead of print. https://doi.org/10.1016/j.lssr.2026.04.012 , May-2026
Articles in Peer-reviewed Journals	Burke M, Wong K, Talyansky Y, Mhatre S, Mitchell C, Juran C, Olsen M, Iyer J, Puukila S, Tahimic C, Christenson, L, Lowe M, Rubinstein L, Shirazi-Fard Y, Sowa M, Alwood JS, Ronca A#, Paul A#. Note: #corresponding authors. "Sexual dimorphism during integrative endocrine and immune responses to ionizing radiation in mice." Sci Rep. 2024 Feb 26;14:7334. https://doi.org/10.1038/s41598-023-33629-7 ; PMID: 38409284; PMCID: PMC10897391 , Feb-2024